COURSE INFORMATION

# GEOSPATIAL REPRESENTATION LEARNING

Course Structure and Format

Winter Semester 2026/27
Mobile Robotics, Geodesy and Geoinformation & Geodetic Engineering

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# Welcome

## Getting to know the group

Short introductions:

- Name and study program
- Previous experience with geospatial data, machine learning, or remote sensing
- One topic or skill you hope to learn in this course

Alternative: start with a short **Wooclap poll** to collect backgrounds, expectations, and questions anonymously.

# Course Learning Outcomes

After completing the course, you will be able to:

1. **Explain** the main concepts of *geospatial representation learning*, including *explicit embeddings*, *embedding databases*, *location encoders*, and *implicit neural representations*.

2. **Identify and compare** relevant *geospatial data sources*, including satellite imagery, time series, GIS layers, and multimodal Earth observation datasets.

3. **Implement and evaluate** selected *machine learning workflows* for geospatial data using *Python-based tools*.

4. **Critically assess** recent *research papers* in geospatial AI, foundation models, and Earth embeddings.

5. **Design and execute** a small *research-oriented geospatial analysis project* in a student group.

6. **Develop and evaluate** a *prototype AI agent* for geospatial data analysis.

7. **Communicate** technical results through *written exercise reports*, *oral presentations*, and *scientific discussion*.

# Three Phases:

### Phase 1: Lectures & Labs

Build conceptual and technical foundations through lectures and hands-on exercises.

**Examination:** final written exam on the course content.

### Phase 2: Paper Analysis

Analyze and review recent research papers in student groups.

**Examination:** pass/fail paper presentation with questions.

### Phase 3: Group Projects

Develop code and conduct a research-oriented geospatial analysis project in groups.

**Examination:** written project report.

**Bloom's cognitive levels across the course**

Remember | Understand | Apply | Analyze | Evaluate | Create

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# Timeline at a Glance

| Course week | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Calendar week** | **CW42** | **CW43** | **CW44** | **CW45** | **CW46** | **CW47** | **CW48** | **CW49** | **CW50** | **CW51** | **CW02** | **CW03** | **CW04** | **CW05** |
| Week starts | Oct 12 | Oct 19 | Oct 26 | Nov 2 | Nov 9 | Nov 16 | Nov 23 | Nov 30 | Dec 7 | Dec 14 | Jan 11 | Jan 18 | Jan 25 | Feb 1 |
| **Phase 1** | **Lectures & Labs** | | | | | | | | | | | | | |
| **Phase 2** | | | | | | | **Paper Analysis** | | | | | | | |
| **Phase 3** | | | | | | | | | | **Group Projects** | | | | |

**University of Bonn academic calendar WS 2026/27:** Lecture period Oct 12, 2026 - Feb 05, 2027 · First-semester welcome Oct 8 · Opening of the academic year Oct 19 · Dies Academicus Dec 2 · Christmas break Dec 24 - Jan 6

## Examination Events

**Phase 1 Written exam**
Date: TBD

**Phase 2 Presentation pass/fail**
Date: TBD

**Phase 3 report deadline**
Date: TBD

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# Phase 1: Lectures and Labs

## Lectures

**1-2 hours**

Presentation slides introduce the core concepts, terminology, and research context.

They provide the shared conceptual foundation for the hands-on work.

## Labs

**Remaining 1-3 hours**

Google Colab notebooks provide hands-on practice:

- Explore and process geospatial datasets
- Implement selected machine learning workflows in Python
- Work with embeddings, retrieval, and location-based representations
- Connect concepts from the lecture to executable examples

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# Phase 1: Content

**Lecture 1**

Why Geospatial Representation Learning?

**Lecture 2**

Geospatial Data: Images, Maps, Time Series

**Lecture 3**

Deep Learning for Geospatial Prediction

**Lecture 4**

Self-Supervised Learning and Foundation Models

**Lecture 5**

Earth Embeddings and Location Encoders

**Lecture 6**

Spatiotemporal Representations and Neural Fields

**Lecture 7**

Geospatial Retrieval, Reasoning, and Agents

**Written Examination**

Phase 1 content

# Phase 2: Paper Analysis

## Author Roleplay

PHD1

PHD2

PROF1

PROF2

PROF3

Each group presents the paper as if they were the author team: two students own the technical details, while three students connect the work to the field and review other groups' papers.

## Roles and Responsibilities

- **PhD Candidate**: Own the paper to a level where you can answer detailed technical questions
- **Professor**: Have an overivew over the paper and its impact in the field and serve as reviewer for other papers

The Thirty-Ninth AAAI Conference on Artificial Intelligence (AAAI-25)

**SatCLIP: Global, General-Purpose Location Embeddings with Satellite Imagery**

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**Abstract**

Geographic information is essential for modeling tasks in fields ranging from ecology to epidemiology. However, extracting relevant location characteristics can be challenging, often requiring expensive data fusion or distillation from massive global imagery datasets. To address this challenge, we introduce Satellite Contrastive Location-Image Pretraining (SatCLIP). This *global, general-purpose geographic location encoder* learns an implicit representation of locations by matching CNN and ViT inferred visual patterns of openly available satellite imagery with their geographic coordinates. The resulting SatCLIP location encoder efficiently summarizes the characteristics of any given location for convenient use in downstream tasks. In our experiments, we use SatCLIP embeddings to improve performance on nine diverse geospatial prediction tasks including temperature prediction, animal recognition, and population density estimation. SatCLIP consistently outperforms alternative location encoders and shows promise for improving geographic domain adaptation. The results demonstrate the potential of vision-location models to learn meaningful representations of our planet from the vast, varied, and largely untapped modalities of geospatial data.

**Introduction**

Satellite imagery has proven to be a valuable source of input data for predictive models across a wide range of real-world applications (Rolf et al. 2024), for example, interpolating missing air pollution data (Chen et al. 2019), crop yield forecasting (Tseng, Kerner, and Rolnick 2022), and agroforestry carbon stock prediction (Reiersen et al. 2022). Patterns extracted from satellite images can describe the unique characteristics of locations, by capturing their natural and built environment. Importantly, characteristics are often correlated in space: While two nearby locations are more likely to have similar features (e.g. the same land cover), two distant locations can also share location characteristics when they share similar environmental ground conditions like climate zones (Fig. 1, right).

Since the spatial patterns governing different geographic data modalities are often complex and non-linear, predictive models working with geo-data benefit from explicitly integrating intuitions for spatial and spatio-temporal dependencies (Klemmer and Neill 2021; Klemmer et al. 2022; Cole





Figure 1: **Motivation for SatCLIP:** Capturing ground conditions from satellite images and transferring them into a location encoder via constrastive image-location pretaining.

et al. 2023). Many geospatial modeling approaches in the sciences even directly leverage geographic location for improving predictions – in fields ranging from epidemiology, the Earth system sciences, to ecology (Mac Aodha, Cole, and Perona 2019; Cole et al. 2023).

But integrating geographic information into a deep learning model is not straightforward. Even though spatial coordinates are often informative, introducing them as features can amplify geographic distribution shift problems and lead to poor evaluation accuracy. This is especially of concern for the setting of geospatial domain adaptation, when data from evaluation areas (and their coordinates) are absent in the training data. On the other hand, models that ignore the spatial nature of the problem will generally perform worse in problems of spatial interpolation, where the evaluation area overlaps with the training area. This brings about an important challenge for representing Earth's diverse surface: capturing the rich information in ground conditions that computer vision models trained on satellite imagery can represent, while simultaneously respecting and integrating the spatial structures and interdependence of geospatial data.

We tackle this research problem by pretraining location encoders: models that take in latitude and longitude, and output a high dimensional embedding of any place on earth. Specifically, we leverage the location information indexing satellite images as an input to a contrastive pretraining objective that aims to match location-image pairs, similarly to the vision-language pretraining deployed in Contrastive Language Image Pretraining (CLIP) (Radford et al.

# Phase 3: Group Project

## Core project idea

Students investigate geospatial questions by **conducting a research study** or **developing GeoAI agents**.

Example themes:

- Deforestation monitoring
- Marine litter monitoring
- Urban expansion
- Agricultural change
- Flood or drought impacts
- Biodiversity and habitat suitability
- Land-cover and land-use change
- Climate-risk indicators
- Infrastructure and mobility patterns
- Human-environment interactions

## Examination Method: Short Research Paper

### Application Paper

An in-depth study of a specific environmental or socio-economic phenomenon using geospatial AI models and representations.

### Method Paper

A methodology paper describing how the group developed an AI agent to dynamically study a specific environmental or socio-economic phenomenon.